why ten?

digitalisierung und programmierung · prof. dr. nicolas meseth

1 the place value trick
2 why two?
3 reading bytes
4 your own base

part 1

the place value trick

why do we count to ten?

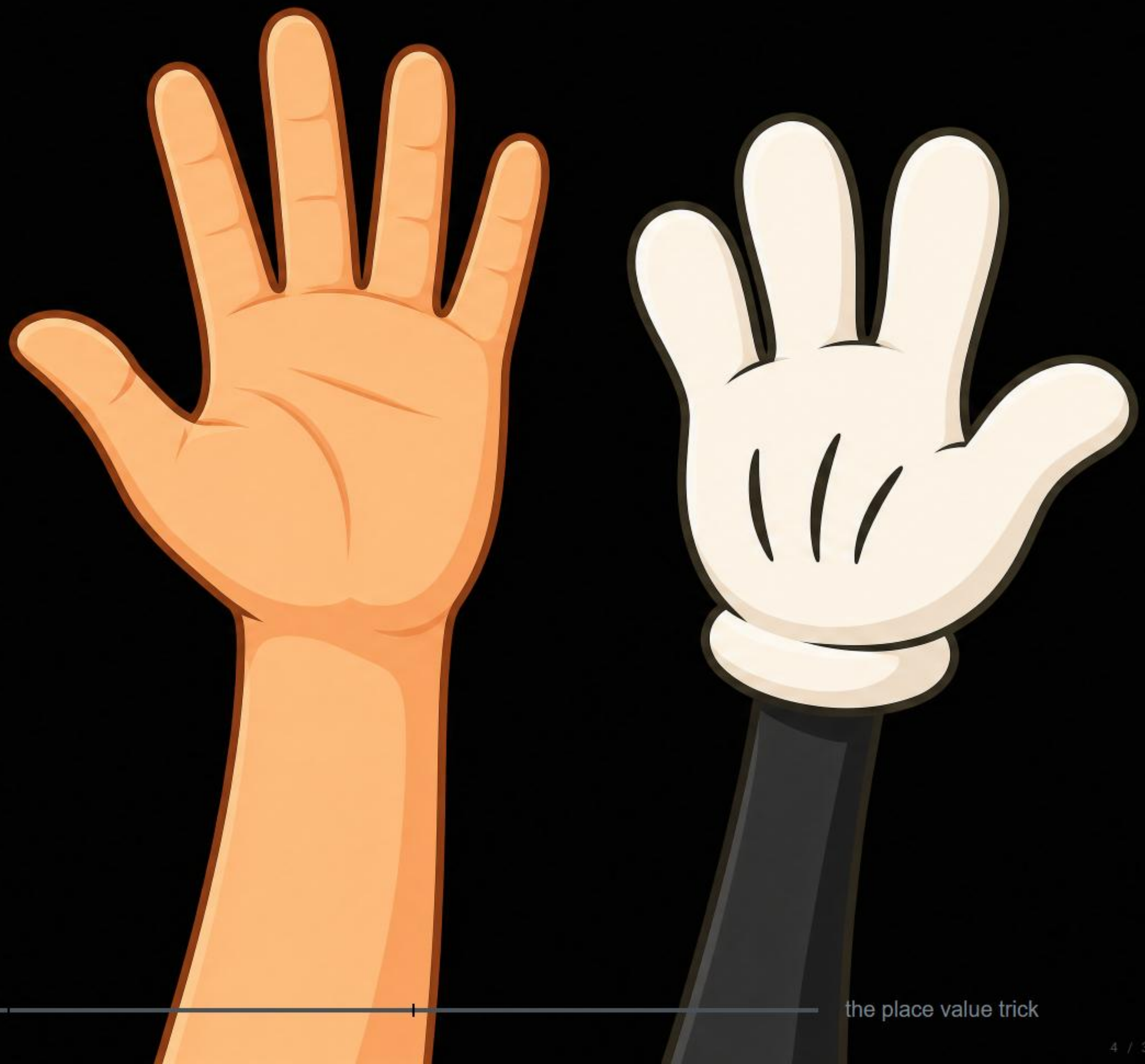


image: ai-generated (gpt-image-2)

what 123 really says

base 10: powers of ten

1	2	3
×	×	×
100	10	1

what 123 really says

base 10: powers of ten

1	2	3
×	×	×
100	10	1

$$1 \cdot 100 + 2 \cdot 10 + 3 \cdot 1 = 123$$

every place carries a power of ten.

the eight-finger creature writes 123

base 8: powers of eight

1	2	3
×	×	×
64	8	1

base 8: powers of eight

1	2	3
×	×	×
64	8	1

$$1 \cdot 64 + 2 \cdot 8 + 3 \cdot 1 = 83$$

same recipe. different base. different number.

and a creature with two flippers?

image: ai-generated (gpt-image-2)

two flippers: counting in binary

we write	the dolphin writes
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000

a new place opens
when the digits run out,
just like 9 to 10

the dolphin writes 110

base 2: powers of two

1	1	0
×	×	×
4	2	1

base 2: powers of two

1	1	0
×	×	×
4	2	1

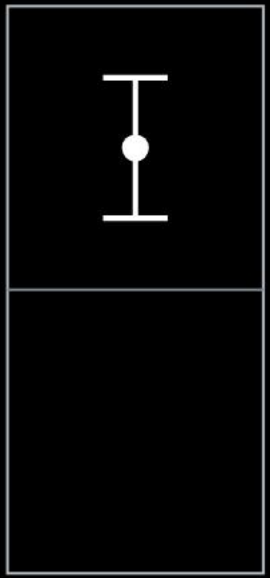
$$1 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 = 6$$

nothing new. the third time is the same trick.

part 2

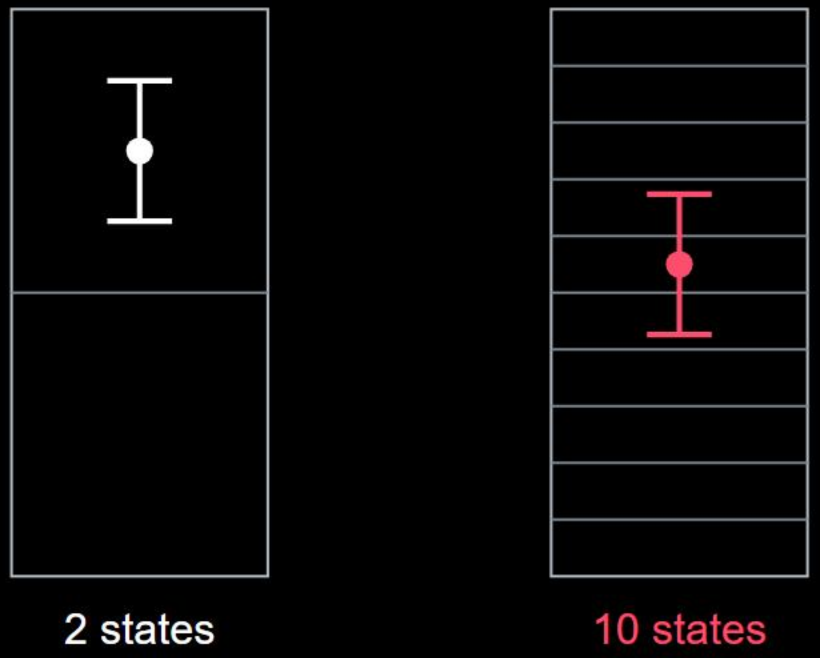
why two?

why two, of all things?



2 states

why two, of all things?



the same wobble on the wire.
with two states it never matters,
with ten it always does.

the price: more places.
places are cheap.

two states are the cheapest thing you can build reliably.

eight bits make a byte

place values

128	64	32	16	8	4	2	1
0	1	0	0	0	0	0	1

eight bits make a byte

place values

128	64	32	16	8	4	2	1
0	1	0	0	0	0	0	1

$$64 + 1 = 65$$

2^8 = 256 values, 0 to 255

256 values, and every one of them is a place value sum.

try it: eight switches, one number

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	1	0	0	0	0	0	1

= 64 + 1 = 65

binary

0100
0001

grouped in two nibbles of four bits

decimal

65

0 to 255

hex

41

one digit per nibble

as text (ascii)

A

character number 65 in the ascii agreement

-1 +1 random reset

part 3

reading bytes

hex: four bits at a time

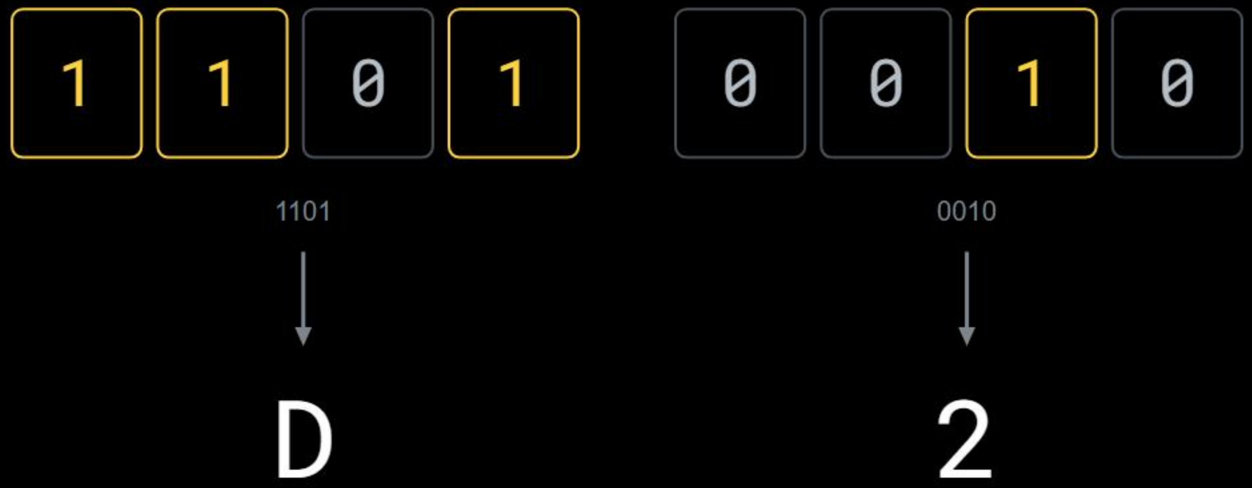
1	1	0	1

1101

0	0	1	0

0010

hex: four bits at a time



four bits have 16 states.
so one hex digit fits a nibble,
and a byte fits two.

hex: four bits at a time

1	1	0	1		0	0	1	0

1101 → D

0010 → 2

four bits have 16 states.
so one hex digit fits a nibble,
and a byte fits two.

ten digits are not enough, so six letters help out

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

hex is not a new number world. it is a reading aid.

try it: the same file, twice

picture: 

pixels: 8 × 8 | 16 × 16 | 32 × 32

what a viewer makes of the bytes

click a pixel

open as text | reset file

the bytes (a hex editor)

```
0000  42 4d 36 03 00 00 00 00 00 00 36 00 00 00 28 00
0010  00 00 10 00 00 00 10 00 00 00 01 00 18 00 00 00
0020  00 00 00 03 00 00 c4 0e 00 00 c4 0e 00 00 00 00
0030  00 00 00 00 00 00 0a 0b 17 0e 11 1c 6b 69 62 45
0040  41 41 11 16 1e 12 19 20 13 19 20 10 18 1e 10 1a
0050  44 1b 22 42 0d 17 82 19 21 b0 1b 25 e4 26 3a f6
0060  28 3d f9 23 36 f9 0c 0d 1a 56 5e 66 a2 a4 a1 38
0070  3a 3e 10 17 1e 12 1c 21 1d 27 31 11 13 21 0b 0b
0080  1d 06 06 18 2e 33 62 28 30 91 1b 1e da 1e 26 e6
0090  26 3a f2 27 37 f5 21 25 32 9e a8 ae a2 a9 ab 55
00a0  57 55 16 17 1a 31 2e 31 3a 33 36 2a 24 27 25 20
00b0  23 1b 16 1f 40 3a 45 a2 94 a8 4b 4d c6 18 12 e0
00c0  17 0c df 12 0b da 50 57 62 9e ab b6 a1 a9 af 6a
00d0  69 68 12 10 11 4d 48 49 2d 27 26 2e 27 27 24 1e
00e0  20 25 1f 23 96 89 8e c5 b5 bf 97 90 c5 22 29 ee
00f0  1d 18 eb 1a 13 e9 83 8f 9c 9d ad bb a2 ad b5 90
0100  8e 88 42 3a 39 4d 45 43 2c 26 25 2c 25 25 2d 26
0110  26 75 6b 6e ca bb be ce bf c5 ca bd c7 72 7f de
```

hover a byte

file header | info header | pixel data | selected pixel

822 bytes = 14 file header + 40 info header + 768 pixel data (16 × 16 pixels × 3 bytes)

photos: ai-generated (gpt-image-2)

two ladders of units

	decimal: each step × 1000	binary: each step × 1024
byte	1	1
kilobyte	1 000	1 024
megabyte	1 000 000	1 048 576
gigabyte	1 000 000 000	1 073 741 824
terabyte	1 000 000 000 000	1 099 511 627 776

both are called the same in everyday speech. that is where the trouble starts.

on the box:
1 TB

1 000 000 000 000 bytes

counted up the decimal ladder, × 1000

on the box:
1 TB

same bytes →

your computer:
931 GB

1 000 000 000 000 bytes

counted up the decimal ladder, × 1000

1 000 000 000 000 ÷ 1 073 741 824

counted down the binary ladder, ÷ 1024^3

nobody cheated. two ladders, one word.

part 4

your own base

you built one already

you built one already

the agreed digits

0

1

2

3

one letter, three colour positions

× 4^2 × 4^1 × 4^0

$$0 \cdot 16 + 3 \cdot 4 + 1 \cdot 1 = 13$$

you built one already

$$0 \cdot 16 + 3 \cdot 4 + 1 \cdot 1 = 13$$

$4^3 = 64$ combinations, enough for an alphabet

k colours are k digits, and position matters.

ten was never special.

ten was never special.

it is the number of fingers on two hands, and nothing else.